Chocolate Sales Performance Analysis

Power BI & Excel | Project Documentation and Business Insights

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1. Project Overview

This project transforms retail chocolate sales transactions into a six-page interactive Power BI report to evaluate revenue, profitability, products, store locations, customer composition, time trends and discount performance. It combines a dimensional data model with DAX measures and business-focused visualizations to make performance comparisons accessible to decision-makers.

The completed report records **revenue of 25,238,648.22 and profit of 10,095,641.91,** representing an approximately 40% margin. Its scope covers 200 products, six brands, five categories, 100 stores and approximately 50,000 customers. The currency symbol is not consistently established by the provided materials, so monetary figures are presented without an assumed currency.

2. Business Questions and Objectives

The analysis addresses how sales and profit are distributed across products and brands; which markets and store formats contribute most; how customers are distributed by age, gender and loyalty status; how performance changes between 2023 and 2024; and how realized revenue, quantity and margin compare across discount bands. The aim is to support targeted investigation and monitoring, rather than claim causal effects from descriptive sales data.

3. Dataset and Analytical Workflow

3.1 Source tables and model

Table	Role in analysis
Sales	Transaction-level fact table: dates, identifiers, quantity, pricing, revenue, cost and profit.
Products	Product dimension: product, category, brand and product attributes.
Stores	Location and channel dimension: store, city, country and store type.
Customers	Customer dimension: demographics, loyalty status and join date.
Calendar	Date dimension for period-based comparisons and time intelligence.

Sales is the central fact table, related to Products, Stores, Customers and Calendar using the corresponding keys. This star-schema-style arrangement supports filtering and aggregation across the report pages.

3.2 Data Preparation and Validation

Excel supported initial inspection; Power Query was used to review data types and identifier consistency and prepare tables for the model. The earlier project notes record that transactions with product IDs unmatched to the Products table were excluded. The exact number of removed records and the final

cleaned row count were not supplied and should be verified directly in the Power BI model before these are reported numerically.

3.3 Measures and Report Design

DAX was used for total revenue, total profit, quantity, orders, unique customers, average order value, profit margin and time-based comparisons. Age groups and discount bands support segmented analysis. The finished report contains Executive Summary, Product Performance, Store and Geography, Customer Insights, Time Analysis, and Discounts, Revenue & Profitability pages, with filtering and drill-down functionality where shown.

4. Results and Key Findings

4.1 Business-wide performance

Indicator	Reported result
Revenue	25,238,648.22
Profit	10,095,641.91
Profit margin	40% (rounded)
Orders	990K
Customers	50K
Units sold	2,970,470 (discount summary total)
Average order value	25.49

Across the dataset, gross reported profit is approximately two-fifths of revenue. The Average order value is the result of dividing the total revenue by the total number of orders.

4.2 Products and Brands

Praline generated the highest category profit, approximately 2.7M, ahead of White (2.4M) and Dark (2.1M). Dark Chocolate 50% ranked first among individual products, with 1.78M in revenue and 709.74K in profit; Truffle Chocolate 80% followed with 1.64M in revenue and 656.76K in profit. Cadbury (4,666,426) and Ferrero (4,689,031) were the leading brands by revenue, *with Ferrero narrowly ahead*. Product-level rankings should guide assortment monitoring and stock investigations, not automatic expansion decisions without demand and inventory context.

4.3 Stores and Geography

Australia contributed approximately 6.319M in revenue, followed by Canada at 4.548M. Toronto was the leading city at approximately 4.5M, followed by London and Paris (around 4.0M each). Chocolate Store 74 ranked first among individual stores, with 258.87K in revenue. Airport locations were the largest store-type segment with 7.54M (29.87% of revenue), while retail locations contributed 4.81M (19.06%). Country and channel rankings reflect the number and composition of stores, and should not be interpreted as per-store productivity without normalization.

4.4 Customer Profile

The dashboard identifies approximately 50,000 customers. Loyalty membership is almost evenly split: 25,104 loyal members (50.21%) and 24,896 non-members (49.79%). The largest age segment is 56+, with 14,260 customers (28.52%). Customer counts are similarly balanced across the displayed gender and loyalty combinations. These visuals describe customer composition; they do not, by themselves, establish that loyalty membership causes higher spend or retention.

4.5 Trends Over Time

The detailed monthly table reports 2023 revenue of 12,612,047.40 and 2024 revenue of 12,626,600.82. The increase is 14,553.42, equivalent to approximately 0.12% year over year. February is the lowest combined-revenue month (1,960,211.41), while January is highest (2,157,534.34). Variation within these two years can be described, but recurring seasonality cannot be established from this short period alone.

4.6 Discount Performance

Transactions with no discount account for 16,724,396.45 in revenue, 6,690,987.45 in profit and 1,857,614 units, or approximately 66.3% of total revenue and 62.5% of units. The 6–10%, 11–15% and 16–20% discount bands contribute 2,993,515.80, 2,850,976.38 and 2,669,759.59 in revenue respectively. Crucially, the summary table displays a 40% profit margin in every discount band; it does not demonstrate margin erosion as discounts rise. Lower absolute revenue at higher discounts does not prove that discounts reduced demand, because band sizes, prices, and underlying product mix may differ.

5. Business Recommendations

- Prioritize monitoring of high-contribution products, particularly Dark Chocolate 50% and Truffle Chocolate 80%, using both profit and sales volume alongside inventory availability.
- Compare markets and store formats on a per-store basis before reallocating resources; Australia and airport stores lead in total revenue but are not necessarily the most productive on a normalized basis.
- Segment customers by actual revenue, frequency and average order value to evaluate whether loyalty status and age groups relate to differences in purchasing behaviour.
- Review the discount-policy and cost-calculation logic, given the uniform 40% margins; assess discount effectiveness using comparable products and periods rather than aggregate band totals alone.
- Validate year-over-year measures, data coverage, and order identifiers, then maintain a consistent KPI definition sheet before publishing the dashboard.

6. Interpretation Limits and Quality Checks

This is descriptive analysis. It does not identify the causes of sales differences, and the source materials do not include sufficient evidence about stockouts, marketing exposure, customer acquisition cost, or operating expenses outside the recorded cost measures. The supplied documentation records exclusions of unmatched product IDs but does not quantify them. The report also shows customer join dates extending into 2025 despite revenue analysis limited to 2023–2024; this should be reviewed for data consistency and time-based customer interpretation.

The Executive Summary and Time Analysis pages show a **0.12% year-over-year revenue growth** from 2023 to 2024, indicating relatively stable revenue performance with a modest increase in sales.

The discount analysis reveals a **consistent profit margin of 40% across all discount bands**, suggesting that the recorded profitability ratio remains stable regardless of discount level. However, products sold without discounts generated the highest total revenue and profit, accounting for the largest share of sales volume.

These findings are descriptive and should not be interpreted as evidence of causal relationships. The analysis does not establish whether discounts directly influence purchasing behaviour, sales volume, or profitability. Additionally, the two-year reporting period limits the ability to identify long-term growth patterns or confirm recurring seasonal trends.

7. Deliverable and Conclusion

The completed deliverable is an interactive six-page Power BI report supported by an Excel/Power Query preparation workflow and a dimensional model. Its strongest contribution is the ability to explore which products, locations, customer groups, and discount segments account for observed sales and profit. Once the highlighted KPI and narrative checks are resolved, the project provides a clear portfolio demonstration of data preparation, modeling, DAX, visualization, and evidence-based interpretation.